Federated Learning Experiments in HAR

From Model Weights to Class Prototypes

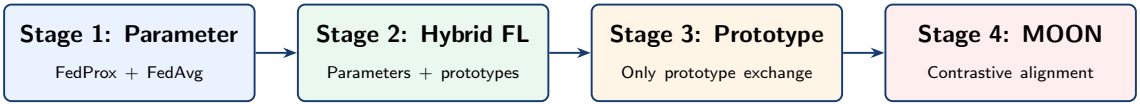
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13 October 2026

Research Question and Project Progression

Central Question

How should distributed clients share learned knowledge when their sensor data and activity distributions differ?



- **Stage 1** shares the model's learned weights.
- **Stage 2** shares weights *and* class-level representation summaries.
- **Stage 3** is the intended parameter-free communication design.
- **Stage 4** returns to parameter sharing but corrects local drift via contrastive feature alignment.

Pipeline Overview

Objective

Developed and compared four federated learning approaches for human activity recognition (HAR), exploring parameter, prototype, and contrastive knowledge sharing.

1. **Dataset Creation (`dataset_creation.py`):** Standardizes WISDM and MotionSense into 6-channel, 60-step sensor windows.
2. **Feature Engineering (`tilt_dataset.py`):** Derives pitch and roll, extending the input to 8 channels.
3. **Stage 1: Parameter-Based FL (`fl_final.py`):** Trains with FedProx and aggregates model parameters using weighted FedAvg.
4. **Stage 2: Hybrid FL (`fl_protofed.py`):** Combines model-parameter aggregation with class-prototype sharing.
5. **Stage 3: Prototype-Based FL (`fl_fedproto.py`):** Uses prototype sharing and regularization without model-parameter aggregation.
6. **Stage 4: Model-Contrastive FL (`fl_moon.py`):** Uses contrastive representation learning alongside parameter aggregation.

Dataset Creation: Sourcing and Standards

Script: `dataset_creation.py`

- **Sources Used:** WISDM (Phone/Watch) and MotionSense.
- **Target Classes:** Maps distinct dataset labels to a unified 4-class set:
  - A: Walking, B: Jogging, D: Sitting, E: Standing.
- **Standardization Parameters:**
  - **Target Frequency**: 20 Hz
  - **Non-overlapping Window Size**: 3 seconds (60 samples per window)
  - **Channels**: 6 (Accelerometer X, Y, Z and Gyroscope X, Y, Z)
  - **3-second window**: timestamp at 50 ms each.
  - **Partitioning**: Data is divided client-disjointly into *Train (70%,* **53 clients***)*, *Validation (15%,* **12 clients***)*, and *Test (15%,* **10 clients***)* to prevent data leakage.

Feature Engineering: Pitch and Roll

Script: `tilt_dataset.py`

- **Objective:** Augment the 6 raw channels with explicit spatial orientation features (Pitch and Roll).
- **Mathematical Derivation:**

$$\text{Pitch} = \tan^{-1}\left(\frac{A_x}{\sqrt{A_y^2 + A_z^2}}\right) \times \left(\frac{180}{\pi}\right)$$

$$\text{Roll} = \tan^{-1}\left(\frac{A_y}{\sqrt{A_x^2 + A_z^2}}\right) \times \left(\frac{180}{\pi}\right)$$

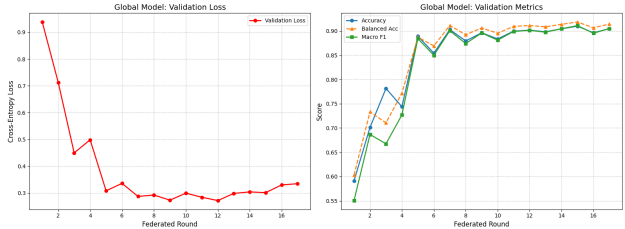
- **Execution:** The script applies vectorized NumPy operations to calculate these angles across all 60 time-steps for every window.
- **Output:** Appends 120 new columns (`pitch_1` to `pitch_60`, `roll_1` to `roll_60`) to the datasets.

Stage 1: Parameter-Based FL

Script: `fl_final.py`

- **Data Preprocessing:** Z-score standardization strictly fitted on the training set.
- **Model Architecture (`OptimizedSensorCNN`):**
  - 1D Convolutional Neural Network processing 8 channels (6 sensor + 2 tilt).
  - Group Normalization and Max Pooling applied after Convolutions.
  - Fully Connected layer outputs a 128-dimensional representation vector, followed by a classification head.
- **Federated Strategy:** FedProx algorithm. Adds a proximal term (μ) to the local objective to restrict models from drifting too far from the global model during local epochs. 25 rounds of FL are performed with 100% participation.
- **Early Stopping:** Training halts if validation loss fails to improve for 5 consecutive rounds, preventing model overfitting and saving computational resources.
- **Hyperparameter Optimization (Optuna):** The script utilizes Optuna to maximize the Macro-F1 score. 10 HPO trials of 5 rounds each are performed.
  - Learning Rate (`lr`): Log-uniform search.
  - Proximal Term (`mu`): Regularization strength for FedProx.
  - CNN Params: Filter sizes (64/128), Kernel sizes (5, 7, 9), Dropout rate.

Stage 1 Results: Learning Behaviour



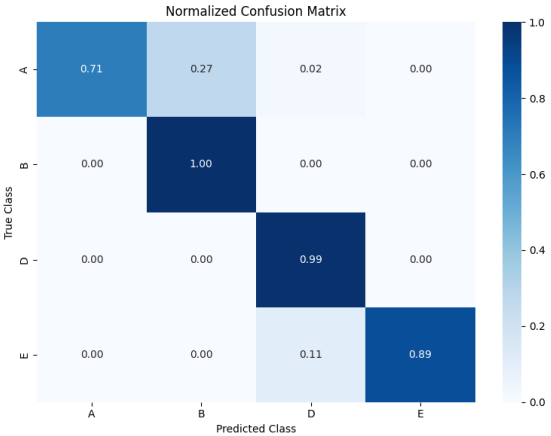
Final classification report: accuracy 88.80%, balanced accuracy 89.64%, Macro-F1 88.58%.

Accuracy

88.80%

- Loss drops sharply during early rounds, then stabilizes around 0.27–0.33.
- Accuracy rises quickly but later rounds show small fluctuations.
- A strong aggregate score does not imply equally strong recall for every class.

Stage 1: Where the Baseline Fails



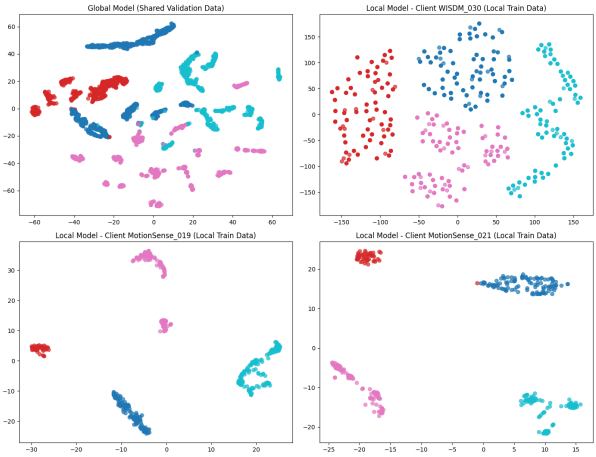
Row-normalized confusion matrix

- Walking (A): 71% correct; 27% predicted as Jogging.
- Jogging (B): 100% correct in this test split.
- Sitting (D): 99% correct.
- Standing (E): 89% correct; 11% predicted as Sitting.

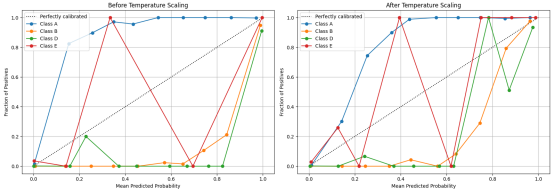
Interpretation

The main errors are asymmetric: Walking is often absorbed into Jogging, while Standing is occasionally confused with Sitting.

Stage 1: Representation and Calibration Diagnostics



t-SNE of learned representations



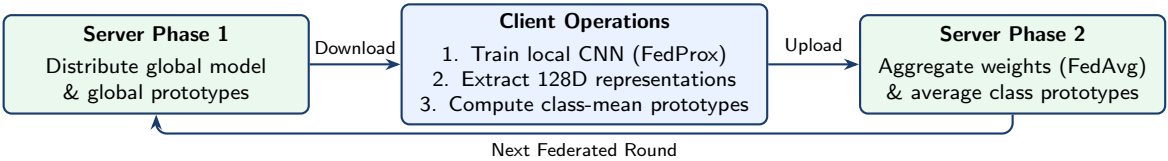
Reliability before/after temperature scaling

Temperature scaling reduced reported ECE from 0.0651 to 0.0339 in the baseline report. t-SNE axes are fitted projections, not meaningful physical coordinates.

Stage 2: Hybrid FL (ProtoFed) – Motivation and Approach

From Parameter-Averaging to Prototype-Inference

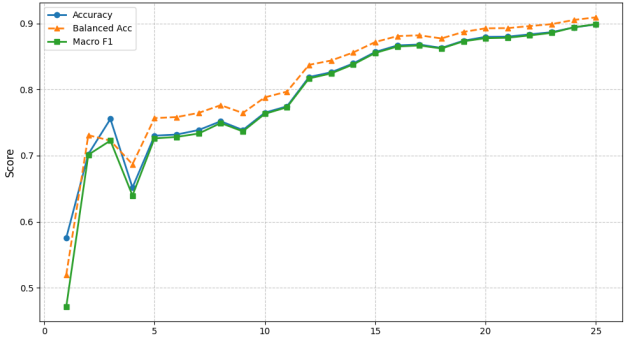
FedProx relied entirely on averaging model weights, including the final linear classification layer. However, averaging classification heads across heterogeneous clients blurs decision boundaries. Stage 2 introduces a hybrid approach: sharing **model parameters** to train feature extractors, but using **class prototypes** for the final classification.

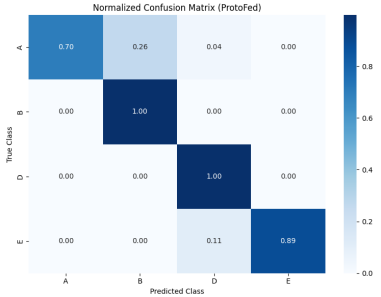


Why Prototype-Based Classification?

During inference, the final linear layer is bypassed completely. Predictions are made by finding the minimum **L2 (Euclidean) distance** between the sample's representation and the global prototypes. This non-parametric approach relies purely on the learned feature space, making it much more robust against the blurry, overlapping boundaries.

Stage 2: ProtoFed – Experimental Results

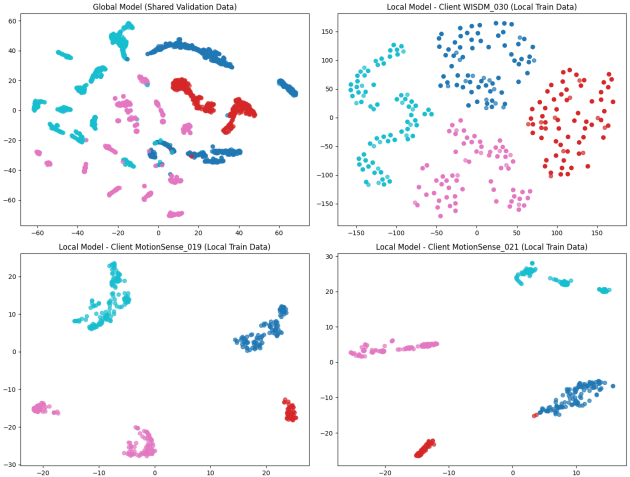




- **Classification report:** Accuracy 88.73%, Balanced Accuracy 89.55%, Macro-F1 88.51%.
- **Strong performance:** Jogging and Sitting recall exceed 99%.
- **Main weakness:** Walking recall is 70.29%; approximately 26% of Walking samples are misclassified as Jogging.
- Around 11% of Standing samples are classified as Sitting.

Prototype sharing provides class-level information, but the Walking–Jogging distinction remains difficult.

Stage 2: ProtoFed – Calibration and Representations



What are we looking at?

This plot shows how the global model groups activities after using ProtoFed. We want to see tight, distinct clusters.

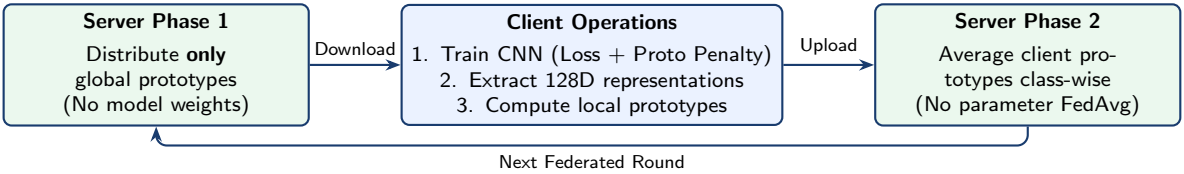
Key Takeaway: Clearer Boundaries

- **Tighter Clusters:** Compared to FedProx, the overlapping zones are smaller because prototype sharing acts as an anchor, forcing clients to align their data spaces better.

Stage 3: Prototype-Only FL (FedProto) – Motivation and Approach

Moving beyond parameter aggregation

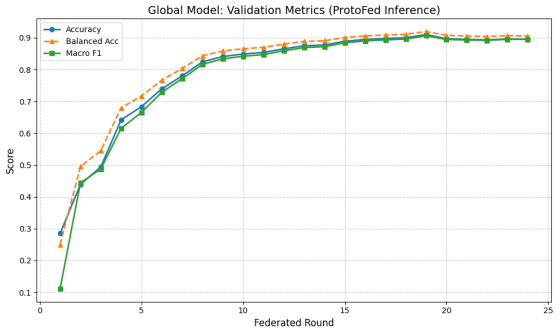
FedProto shares class-level prototypes rather than averaging model parameters at the server. Clients train locally while prototype regularization encourages representations to align with global class prototypes.

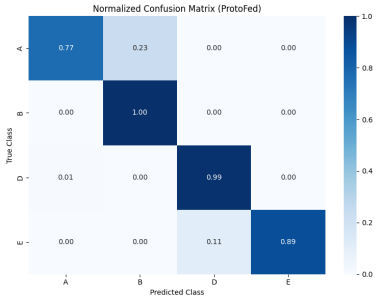


Key Distinction & Inference

No model-parameter aggregation is performed. Inference is done by assigning the class of the nearest global prototype using Euclidean distance. Prototype alignment ensures different local representation spaces become compatible over time.

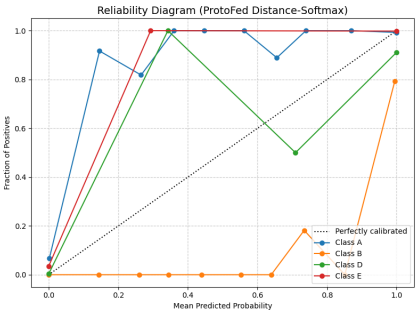
Stage 3: FedProto – Experimental Results

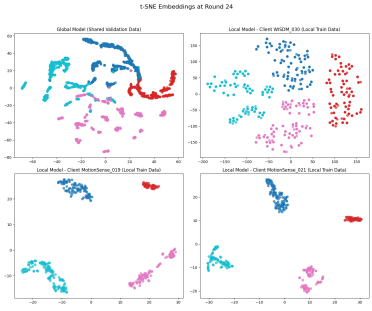




- **Classification report:** Accuracy 90.39%, Balanced Accuracy 91.05%, Macro-F1 90.22%.
- **Strong performance:** Jogging recall is 100%, while Sitting recall is 99.02%.
- **Walking improves:** Recall rises to 76.62%, compared with 70.29% in the ProtoFed classification report. Approximately 11% of Standing samples are still misclassified as Sitting.
- FedProto has **stronger reported classification metrics** than ProtoFed and FedProx.

Stage 3: FedProto – Calibration and Representations



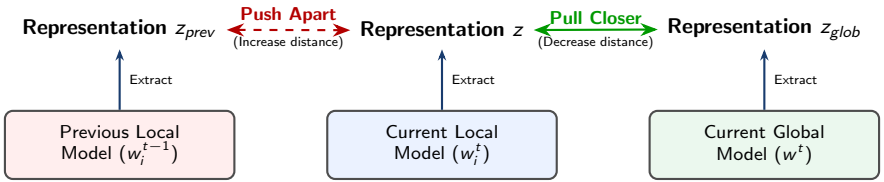


- **Round 24:** Accuracy 89.52%, Balanced Accuracy 90.57%, Macro-F1 89.49%.
- **Calibration:** ECE 0.0878, MCE 0.7000, Brier Score 0.1738.
- Prototype-distance pseudo-probabilities remain insufficiently calibrated.
- t-SNE supports qualitative inspection of class structure and client-specific representations.

Stage 4: Model-Contrastive FL (MOON) – Motivation and Approach

Contrastive Learning at the Model Level

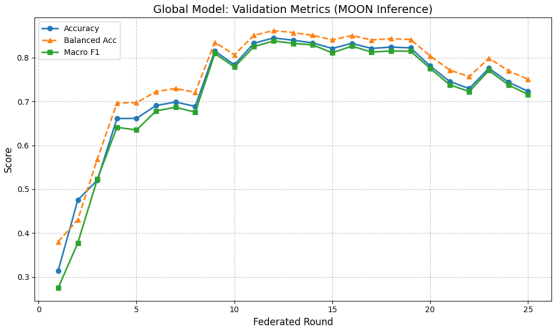
Instead of sharing class prototypes or restricting parameter drift via L_2 norms, MOON leverages **contrastive learning** on the feature representations. The local objective forces the client's representation to align with the global model (positive pair) and diverge from its own previous local model (negative pair).

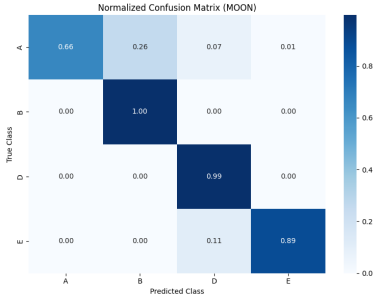


Key Distinction

Like FedProx, MOON communicates **only model parameters** (no prototypes).

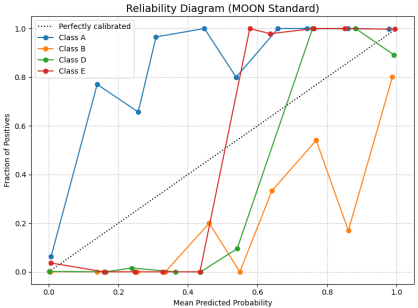
Stage 4: MOON – Experimental Results

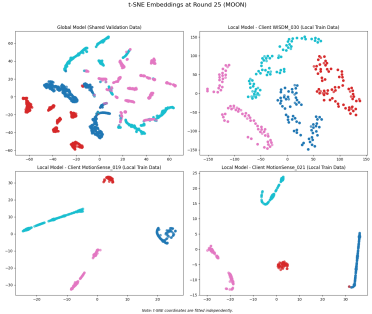




- **Classification report:** Accuracy 87.57%, Balanced Accuracy 88.52%, Macro-F1 87.31%.
- **Performance shift:** While Sitting and Jogging remain highly accurate (>99%), Walking recall drops to **66.30%**, underperforming the FedProx baseline.
- **Observation:** MOON struggles to surpass the standard FedProx baseline in this specific HAR setup.

Stage 4: MOON – Calibration and Representations





- **Calibration Metrics:** ECE 0.0985, MCE 0.3968, Brier Score 0.2115.
- **Reliability:** MOON uses standard output logits for inference (not distances). However, the calibration is slightly degraded compared to the post-temperature scaled FedProx baseline.
- **Feature Space:** The t-SNE plot shows that while static activities cluster well, the overlapping boundaries between dynamic activities (Walking vs. Jogging) remain somewhat tangled despite the contrastive penalty.

Comprehensive Comparison: Aggregation vs. Performance

Algorithm Summary on 4-Class HAR

This table summarizes the final global model performance across the four implemented federated learning strategies using identical client data splits.

Algorithm	**Aggregation Mechanism**	**Knowledge Exchanged**	**Acc.**	**B-Acc.**	**Macro-F1**
FedProx	Weighted FedAvg	Model Parameters	88.80%	89.64%	88.58%
ProtoFed	FedAvg + Prototype Avg	Parameters + Prototypes	88.73%	89.55%	88.51%
MOON	Weighted FedAvg	Model Parameters	87.57%	88.52%	87.31%
FedProto	**Prototype Avg**	**Class Prototypes Only**	**90.39%**	**91.05%**	**90.22%**

Final Conclusion

FedProto (Prototype-Only FL) achieved the highest performance and best class separation. The results indicate that exchanging abstract, class-level prototypes is significantly more robust against sensor data heterogeneity than averaging deep model parameters.

References

1. Brendan McMahan, Eider Moore, Daniel Ramage, Seth Hampson, and Blaise Agüera y Arcas, "Communication-Efficient Learning of Deep Networks from Decentralized Data," AISTATS, 2017.
2. Tian Li, Anit Kumar Sahu, Manzil Zaheer, Maziar Sanjabi, Ameet Talwalkar, and Virginia Smith, "Federated Optimization in Heterogeneous Networks," Proceedings of Machine Learning and Systems (MLSys), 2020.
3. Yu Qiao, Seong-Bae Park, Sun Moo Kang, and Choong Seon Hong, "Prototype Helps Federated Learning: Towards Faster Convergence," arXiv preprint arXiv:2303.12296, 2023.
4. Yue Tan, Guodong Long, Lu Liu, Tianyi Zhou, Qinghua Lu, Jing Jiang, and Chengqi Zhang, "FedProto: Federated Prototype Learning across Heterogeneous Clients," AAAI, 2022.
5. Qinbin Li, Bingsheng He, and Dawn Song, "Model-Contrastive Federated Learning," CVPR, 2021.

Thank You